

Honours Analysis 1 Tutorials (Sec. 003)

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Please note that this document has not been proofread and may contain typos. Feel free to see or email me if you have any questions.

REMINDER: There are people available at the math helpdesk (Burnside 911) to help answer questions you might have. Also, please see the end of this document for my office hours.

1 Fifth Tutorial

We first recall some notions about open and closed subsets of $\mathbb{R}$. A set $U \subseteq \mathbb{R}$ is called **open** if, for every $x \in U$, there exists $\varepsilon > 0$ such that $V_\varepsilon := (x - \varepsilon, x + \varepsilon) \subseteq U$. In symbolic terms, U is said to be open if and only if:

$$(\forall x \in U) (\exists \varepsilon > 0 \text{ such that } V_\varepsilon(x) \subseteq U).$$

In other words, a set U is open if for every point $x \in U$ there exists an open interval $(x - \varepsilon, x + \varepsilon)$ contained in U .¹ Since an open interval $(x - \varepsilon, x + \varepsilon)$ contains all points “near x ”, the definition for an open set can be interpreted as follows:

A set $U \subseteq \mathbb{R}$ is called open if, for every $x \in U$, the set U also contains all points near x .²

A set $F \subseteq \mathbb{R}$ is said to be **closed** if $F^c = \mathbb{R} \setminus F$ is open in $\mathbb{R}$. Let us now summarize some properties of open sets that you will have proven in the lectures.

¹Note that the empty set $\emptyset$ is vacuously open by this definition as well. If $\emptyset$ were *not* open, then there must exist $x \in \emptyset$ such that $V_\varepsilon(x) \not\subseteq \emptyset$ for all $\varepsilon > 0$. Since this would imply $x \in \emptyset$, we have a contradiction.

²Please note that this is not rigorous, and is only intended to provide intuition.

- If $\{U_\alpha\}_{\alpha \in I}$ is an indexed family of open sets, then $\bigcup_{\alpha \in I} U_\alpha$ is also open.
- If U_1 and U_2 are open subsets of $\mathbb{R}$, then so is $U_1 \cap U_2$. By induction, finite intersections of open sets are thus closed.
- Both $\mathbb{R}$ and $\emptyset$ are open in $\mathbb{R}$.

The set $\mathcal{T}$ of all open subsets of $\mathbb{R}$ is therefore closed under unions, finite intersections, and contains both $\emptyset$ and $\mathbb{R}$. This makes $\mathcal{T}$ into what we call a *topology*. Since $\mathbb{R}$ has a topology (i.e. open subsets), it is called a *topological space*.

Example 1. Let us show that infinite intersections of open sets need not be open. For every $n \in \mathbb{N}$, let U_n be the open interval:

$$U_n := \left(-\frac{1}{n}, \frac{1}{n}\right).$$

Now consider the infinite family $\{U_n\}_{n=1}^\infty$, what can be said about $\bigcap_{n=1}^\infty U_n$? Well, after a moment's consideration, it is not difficult to see that

$$\bigcap_{n=1}^\infty U_n = \bigcap_{n=1}^\infty \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}. \quad (1.1)$$

To prove this, we first note the obvious inclusion $\{0\} \subseteq U_n$, for each $n \in \mathbb{N}$. Since this holds for all n , we have $\{0\} \subseteq \bigcap_{n=1}^\infty U_n$. Conversely, let $x \in \bigcap_{n=1}^\infty U_n$ be given; we will show that $x \neq 0$ is impossible. If $x \neq 0$, then either $x < 0$ or $x > 0$.

1. Case $x > 0$. By the Archimedean property, there exists $n \in \mathbb{N}$ so large that $\frac{1}{n} < x$. Thus, $x \notin U_n$.
2. Case $x < 0$. Then, $-x > 0$. As above, we can find $n \in \mathbb{N}$ such that $\frac{1}{n} > -x$. This implies that $x < -\frac{1}{n}$ whence $x \notin U_n$.

In either case, we get $x \notin U_n$ for some $n \in \mathbb{N}$. Since this contradicts the assumption that $x \in \bigcap_{n=1}^\infty U_n$, we conclude that $x \neq 0$ is impossible. Thus, $x \in \bigcap_{n=1}^\infty U_n$ implies $x = 0$, i.e. $\bigcap_{n=1}^\infty U_n \subseteq \{0\}$. To summarize, we have proven (1.1).

But $\{0\}$ is not open! If it were, then for $0 \in \{0\}$ we could find $\varepsilon > 0$ such that $(0 - \varepsilon, 0 + \varepsilon) \subseteq \{0\}$; which is clearly absurd. Hence $\bigcap_{n=1}^\infty U_n$ is an infinite intersection of open sets that is not open.

Example 2. You have proven that any intersection of closed sets is necessarily closed. You have also seen that finite unions of closed sets are closed. What about infinite unions of closed sets? First, note that any singleton $\{x\} \subset \mathbb{R}$ is closed. Indeed,

$$\mathbb{R} \setminus \{x\} = (-\infty, x) \cup (x, \infty)$$

is a union of open sets, and hence open. Consider the union $\bigcup_{x \in (0,1)} \{x\} = (0, 1)$ of infinitely many closed sets. We claim that $(0, 1)$ is not closed in $\mathbb{R}$. If it were, then $\mathbb{R} \setminus (0, 1) = (-\infty, 0] \cup [1, \infty)$ would be open in $\mathbb{R}$. But, for any $\varepsilon > 0$, the set

$$V_\varepsilon(0) = (-\varepsilon, \varepsilon) \not\subseteq (-\infty, 0] \cup [1, \infty).$$

This means that $\mathbb{R} \setminus (0, 1)$ is not open whence $(0, 1)$ is not closed.

By way of another example, we show that $\mathbb{N}$ is closed in $\mathbb{R}$. Indeed, $\mathbb{R} \setminus \mathbb{N}$ can be written as the union of open sets:

$$\mathbb{R} \setminus \mathbb{N} = (-\infty, 1) \cup \bigcup_{n=1}^{\infty} (n, n+1).$$

Thus, $\mathbb{R} \setminus \mathbb{N}$ is open. A more interesting example is the following.

Example 3. We show that $\mathbb{Q}$ is neither open nor closed in $\mathbb{R}$. Suppose for a contradiction that $\mathbb{Q}$ is open in $\mathbb{R}$. Then, for any $q \in \mathbb{Q}$ there exists $\varepsilon > 0$ such that $V_\varepsilon(q) = (q-\varepsilon, q+\varepsilon) \subseteq \mathbb{Q}$. Recall from the previous tutorial that the irrationals are dense in $\mathbb{R}$. Thus, every open interval in $\mathbb{R}$ contains an irrational number. In particular, so does $V_\varepsilon(q)$. It follows that $V_\varepsilon(q)$ cannot be contained in $\mathbb{Q}$. We conclude that $\mathbb{Q}$ is *not* open.

To see that $\mathbb{Q}$ is not closed, we argue by contradiction. If $\mathbb{Q}$ were closed, then the set of irrationals $\mathbb{R} \setminus \mathbb{Q}$ would be open in $\mathbb{R}$. Then, for any irrational number ξ , we could find $\varepsilon > 0$ such that $V_\varepsilon(\xi) = (\xi - \varepsilon, \xi + \varepsilon) \subseteq \mathbb{R} \setminus \mathbb{Q}$. Since any open interval contains a rational number ($\mathbb{Q}$ is dense in $\mathbb{R}$), this is impossible. Hence, $\mathbb{Q}$ cannot be closed.

1.1 About Intervals

Proposition 1. *Let $U \subseteq \mathbb{R}$. Then U is open if and only if it can be written as a union of open intervals.*

Proof. Since open intervals are open, and unions of open sets are always open, the direction “ $\Leftarrow$ ” is immediate. Conversely, let U be an open subset of $\mathbb{R}$. If $U = \emptyset$, then U can be written as the *empty* union of open intervals. Otherwise, U is non-empty. For every $x \in U$, we can find $\varepsilon_x > 0$ such that $V_{\varepsilon_x}(x) \subseteq U$. Therefore,

$$\bigcup_{x \in U} V_{\varepsilon_x}(x) \subseteq U.$$

On the other hand, $x \in U$ implies $x \in V_{\varepsilon_x}(x)$ from which we get

$$\bigcup_{x \in U} V_{\varepsilon_x}(x) \supseteq U.$$

We conclude that

$$\bigcup_{x \in U} V_{\varepsilon_x}(x) = U.$$

Since every $V_{\varepsilon_x}(x)$ is itself an open interval, we see that U can be written as the union of open intervals. $\square$

In the previous tutorial we proved the characterization of intervals theorem, which we recall below for the sake of convenience.

Theorem 1.1 (Characterization of Intervals). *Let $S \subseteq \mathbb{R}$ contain at least two points and assume that S satisfies the property*

$$\text{if } x, y \in S \text{ and } x < y \text{ then } [x, y] \subseteq S.$$

Then, S is an interval.

Using this theorem, we will be able to deduce a very intuitive result that, a priori, seems *very* tricky to prove rigorously.

Proposition 2. *Let J be a non-empty index set and $\{I_j\}_{j \in J}$ a family of non-empty intervals such that $\bigcap_{j \in J} I_j \neq \emptyset$. Then, $I := \bigcup_{j \in J} I_j$ is an interval in $\mathbb{R}$.*

Proof. As mentioned, we will be using the characterization of intervals theorem. Since every interval contains at least two points, so must I . Let now $x, y \in I$ with $x < y$ be given. It is enough to show that $[x, y] \subseteq I$. Let $a \in \bigcap_{j \in J} I_j$; choose indices j_1 and $j_2 \in J$ such that $x \in I_{j_1}$ and $y \in I_{j_2}$. We now distinguish three cases.

1. Case $a \leq x < y$. Since $a, y \in I_{j_2}$, we must have $[a, y] \subseteq I_{j_2} \subseteq I$. In particular, $[x, y] \subseteq I$.

2. Case $x < a < y$. Since $x \in I_{j_1}$ and $a \in I_{j_1}$, we get that $[x, a] \subseteq I_{j_1} \subseteq I$. Similarly, we see that $[a, y] \subseteq I$. Therefore, $[x, y] = [x, a] \cup [a, y] \subseteq I$.
3. Case $x < y \leq a$. Since $x \in I_{j_1}$ and $a \in I_{j_1}$, it follows from the properties of intervals that $[x, a] \subseteq I_{j_1} \subseteq I$. However, because $y \leq a$, we have

$$[x, y] \subseteq [x, a] \subseteq I.$$

In either case we have $[x, y] \subseteq I$. By the characterization of intervals theorem, I is an interval in $\mathbb{R}$. With this, the proof is complete. $\square$

1.2 The Boundary of a Set

Let X be a subset of $\mathbb{R}$. The *boundary* of X is defined to be the set

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0, V_\varepsilon(x) \cap X \neq \emptyset \text{ and } V_\varepsilon(x) \cap X^c \neq \emptyset\}.$$

In other words, ∂X consists of all points $x \in X$ such that every ε -neighbourhood of x intersects both X and X^c . This definition is a lot to unpack so let us take a step back. Consider the set $(0, 1) \subset \mathbb{R}$. The point $0 \notin (0, 1)$ belongs to $\partial(0, 1)$ since every ε -neighbourhood $(-\varepsilon, \varepsilon)$ of 0 contains negative numbers (points not in $(0, 1)$) and points between 0 and 1 (i.e. points in $(0, 1)$). Similarly, $1 \in \partial S$.

In the tutorial, I will draw a picture that I hope will make of this clearer. If this doesn't help, come see me or send me an email; I will then update these notes with more examples. For now, we will stick to more "proofy" problems.

Proposition 3. *For any subset $X \subseteq \mathbb{R}$, we have the $\partial X = \partial(\mathbb{R} \setminus X)$. Therefore, the boundary of a set is complement symmetric.*

Proof. We could go through showing that $\partial X \subseteq \partial(\mathbb{R} \setminus X)$ and $\partial(\mathbb{R} \setminus X) \subseteq \partial X$. Or, we could be clever and realize that the definition of ∂X is symmetric in X and $\mathbb{R} \setminus X = X^c$. Indeed, the proof follows from the following observation:

$$\begin{aligned} \partial X &= \{x \in \mathbb{R} : \forall \varepsilon > 0, V_\varepsilon(x) \cap X \neq \emptyset \text{ and } V_\varepsilon(x) \cap X^c \neq \emptyset\} \\ &= \{x \in \mathbb{R} : \forall \varepsilon > 0, V_\varepsilon(x) \cap X^c \neq \emptyset \text{ and } V_\varepsilon(x) \cap X \neq \emptyset\} \\ &= \partial(X^c) \\ &= \partial(\mathbb{R} \setminus X). \end{aligned}$$

$\square$

Proposition 4. Let $\emptyset \subsetneq X \subsetneq \mathbb{R}$ be given. Then, $\partial X \neq \emptyset$.

Proof. Since X is non-empty, we can choose $x \in X$. Since $X \neq \mathbb{R}$, there exists $y \in \mathbb{R}$ with $y \notin X$. For the moment, let us make the assumption:

$$\boxed{\text{Assumption : } x < y}$$

Then, $[x, y] \cap X$ is a non-empty and bounded subset of $\mathbb{R}$. Let s be the supremum of $[x, y] \cap X$ in $\mathbb{R}$. Clearly, $x \leq s \leq y$. We now show that $s \in \partial X$. Given $\varepsilon > 0$, we know that $s - \varepsilon$ is not an upper bound of $[x, y] \cap X$. Therefore, we can find an element $z \in [x, y] \cap X$ such that

$$s - \varepsilon < z \leq s.$$

Then, $z \in V_\varepsilon(s) \cap X$. In particular, $V_\varepsilon(s) \cap X \neq \emptyset$. We now consider two sub-cases:

- (i) If $s \notin X$ then $V_\varepsilon(s) \ni s$ intersects $\mathbb{R} \setminus X$ at s .
- (ii) Suppose that $s \in X$ so that $s \in [x, y] \cap X$. Since $y \notin X$ and $s \leq y$, we must have $s < y$. In this case, we obtain

$$(s, \min \{s + \varepsilon, y\}) \subseteq (s, s + \varepsilon) \subseteq V_\varepsilon(s).$$

But, since s is an upper bound for $[x, y] \cap X$, the interval $(s, \min \{s + \varepsilon, y\})$ is contained in $\mathbb{R} \setminus X$. Indeed, for any $w \in (s, \min \{s + \varepsilon, y\})$ we have $x \leq s < w \leq y$ whence $w \in [x, y]$. If $w \in X$, then $w \in [x, y] \cap X$ would contradict the upper bound property of s .

In either case, we have $V_\varepsilon(s) \cap X \neq \emptyset$ and $V_\varepsilon(s) \cap (\mathbb{R} \setminus X) \neq \emptyset$. Thus, $s \in \partial X$. In the case $y < x$, we argue similarly but using instead the infimum of $[y, x] \cap X$. For the full details, see the Assignment 4 solutions on MyCourses. $\square$

We now give an important fact about closed sets.

Lemma 5. Let $X \subseteq \mathbb{R}$ be closed. Then, $\partial X \subseteq X$.

Proof. In logical language, this is equivalent to the statement:

$$x \in \partial X \implies x \in X.$$

Again, the above is equivalent to its contrapositive:

$$x \notin X \implies x \notin \partial X$$

which is also equivalent to the statement:

$$x \in \mathbb{R} \setminus X \implies x \notin \partial X.$$

Thus, it is enough to prove this last implication. Since X is closed, $\mathbb{R} \setminus X$ is open in $\mathbb{R}$. If $x \in \mathbb{R} \setminus X$, there must exist $\varepsilon > 0$ so small that

$$V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \subseteq \mathbb{R} \setminus X.$$

In particular, $V_\varepsilon(x) \cap X = \emptyset$ whence $x \notin \partial X$. □

Remark 1.1. The converse is also true, i.e. if $\partial X \subseteq X$ then X must be closed. To see this, suppose that $\partial X \subseteq X$; we will show that X^c is open in $\mathbb{R}$. Let now $x \in X^c$ be given. If $x \in \partial X$, it would follow that $x \in X$. Thus, we have $x \notin \partial X$. This is precisely the statement that

$$V_\varepsilon(x) \cap X = \emptyset \quad \text{or} \quad V_\varepsilon \cap X^c = \emptyset,$$

for *some* $\varepsilon > 0$. Since $x \in V_\varepsilon(x) \cap X^c$, the only viable option is $V_\varepsilon(x) \cap X = \emptyset$. It follows that $V_\varepsilon(x) \subseteq X^c$. This shows that X^c is open.

A subset X of $\mathbb{R}$ is said to be **clopen** if it is both open and closed. With the help of this lemma, we can completely characterize the clopen subsets of $\mathbb{R}$.

Proposition 6. *A subset X of $\mathbb{R}$ is clopen if and only if $X = \mathbb{R}$ or $X = \emptyset$.*

Proof. It was shown in class that $\emptyset$ and $\mathbb{R}$ are clopen. Thus, we only need to show the “ $\implies$ ” implication. To this end, let X be a non-empty clopen subset of $\mathbb{R}$. We claim that $X = \mathbb{R}$. Since X is closed, our previous lemma forces $\partial X \subseteq X$.

We now show that ∂X must be empty. If this is true, then Proposition 4 would imply that $X = \mathbb{R}$, since $X \neq \emptyset$. Suppose that ∂X is non-empty and choose $x \in \partial X$. As mentioned above, $\partial X \subseteq X$ implies $x \in X$. Since X is open, there exists an ε -neighbourhood $V_\varepsilon(x) \subseteq X$. Thus, $V_\varepsilon(x) \cap (\mathbb{R} \setminus X) = \emptyset$ for this particular ε . Of course, this contradicts the assumption that $x \in \partial X$. Our only option is therefore $\partial X = \emptyset$ which forces (by Proposition 4) $X = \mathbb{R}$. □

From this, we can deduce a very nice topological property.

Corollary 1.2. *$\mathbb{R}$ cannot be written as the union of two non-empty open subsets X and Y of $\mathbb{R}$ such that $X \cap Y = \emptyset$.*

Proof. Let X and Y be open subsets of $\mathbb{R}$, with $X \cap Y = \emptyset$, such that $\mathbb{R} = X \cup Y$. Clearly, $X^c = Y$ and $Y^c = X$. Since Y is open, this tells us that X^c is open. Hence, X is closed. Similarly, Y can be seen to be closed. But, this would mean that X and Y are clopen. By the last proposition, we must have $X = \mathbb{R}$ or $Y = \mathbb{R}$. Without loss of generality, suppose that $X = \mathbb{R}$. Finally, observe that

$$Y = Y \cap \mathbb{R} = Y \cap X = \emptyset.$$

This shows that X and Y cannot be non-empty disjoint open subsets of $\mathbb{R}$. With this, the proof is complete. $\square$

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